

Automobile Imports Price Prediction

PRCP-1017 — Machine Learning Project Report

Dataset: UCI Automobile / Auto Imports (1985)

1. Executive Summary

This project builds a supervised regression model to predict automobile list price from technical specifications (engine size, dimensions, weight, fuel economy) and categorical attributes (make, body style, drive configuration). Ten regression algorithms were trained through an identical preprocessing pipeline and compared using 5-fold cross-validated R^2 as the primary ranking metric, since the 41-row held-out test split is small and noisy.

Gradient Boosting was the top-ranked model (CV $R^2 = 0.874$), narrowly ahead of Ridge and Linear Regression, and clearly ahead of KNN and Decision Tree. After hyperparameter tuning via GridSearchCV, cross-validated R^2 improved to 0.879, with tuned test $R^2 = 0.976$ and test MAE = \$988.70. Vehicle size and powertrain specifications — engine size and curb weight above all — are the dominant price drivers.

2. Dataset

Source: 1985 Auto Imports Database, UCI Machine Learning Repository — 201 vehicles across 22 makes, 26 attributes spanning risk rating, insurance losses, physical dimensions, engine/drivetrain specification, fuel economy, and price (target).

2.1 Missing data

Missing values were encoded in the raw file as the literal string "?" rather than a blank. The following columns had missing entries after conversion to NaN:

Column	Missing
normalized_losses	37 (18.4%)
bore	4 (2.0%)
stroke	4 (2.0%)
num_of_doors	2 (1.0%)
horsepower	2 (1.0%)
peak_rpm	2 (1.0%)

Numeric predictors were later imputed with the column median and categorical predictors with the most frequent category, inside the modeling pipeline (fit on the training split only).

3. Exploratory Data Analysis

3.1 Target distribution

Price is right-skewed (skewness = 1.81) with a long tail from luxury makes such as Jaguar, Porsche, Mercedes-Benz, and BMW.

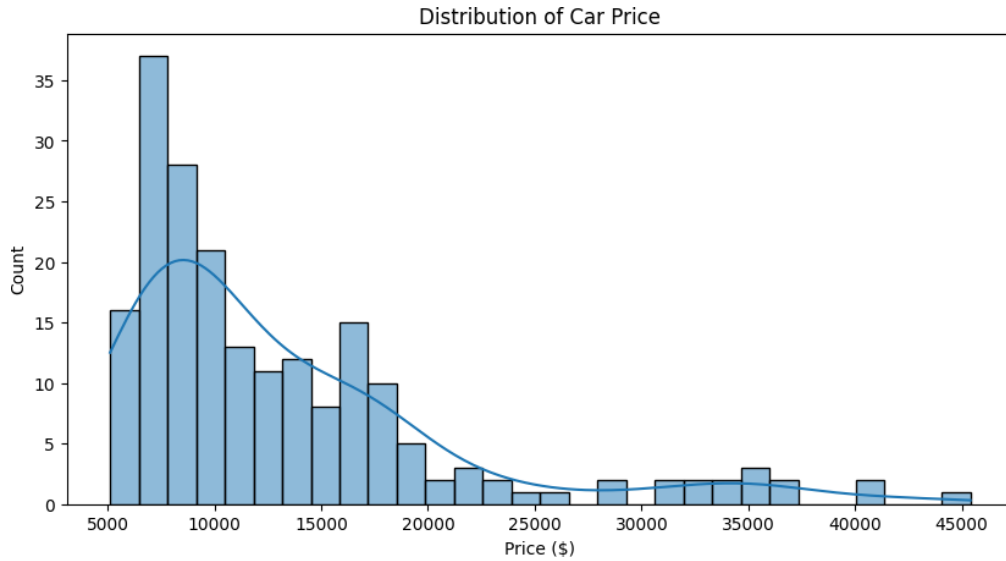


Figure 1. Distribution of car price.

3.2 Correlation structure

Engine size ($r = 0.87$), curb weight ($r = 0.83$), horsepower ($r = 0.81$), and body width/length correlate strongly and positively with price; city and highway MPG correlate strongly and negatively ($r \approx -0.69$ to -0.70) — larger, more powerful cars cost more and use more fuel.

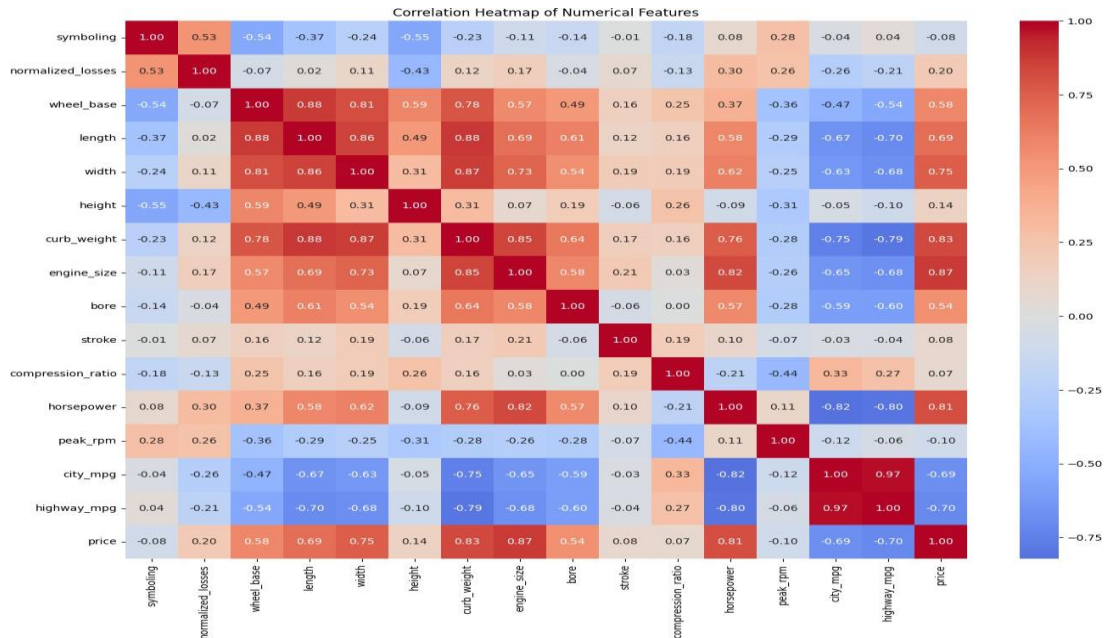


Figure 2. Correlation heatmap of numeric features.

3.3 Price by category

Convertibles and hardtops, rear-wheel drive, diesel, and turbocharged configurations command a price premium over their counterparts.

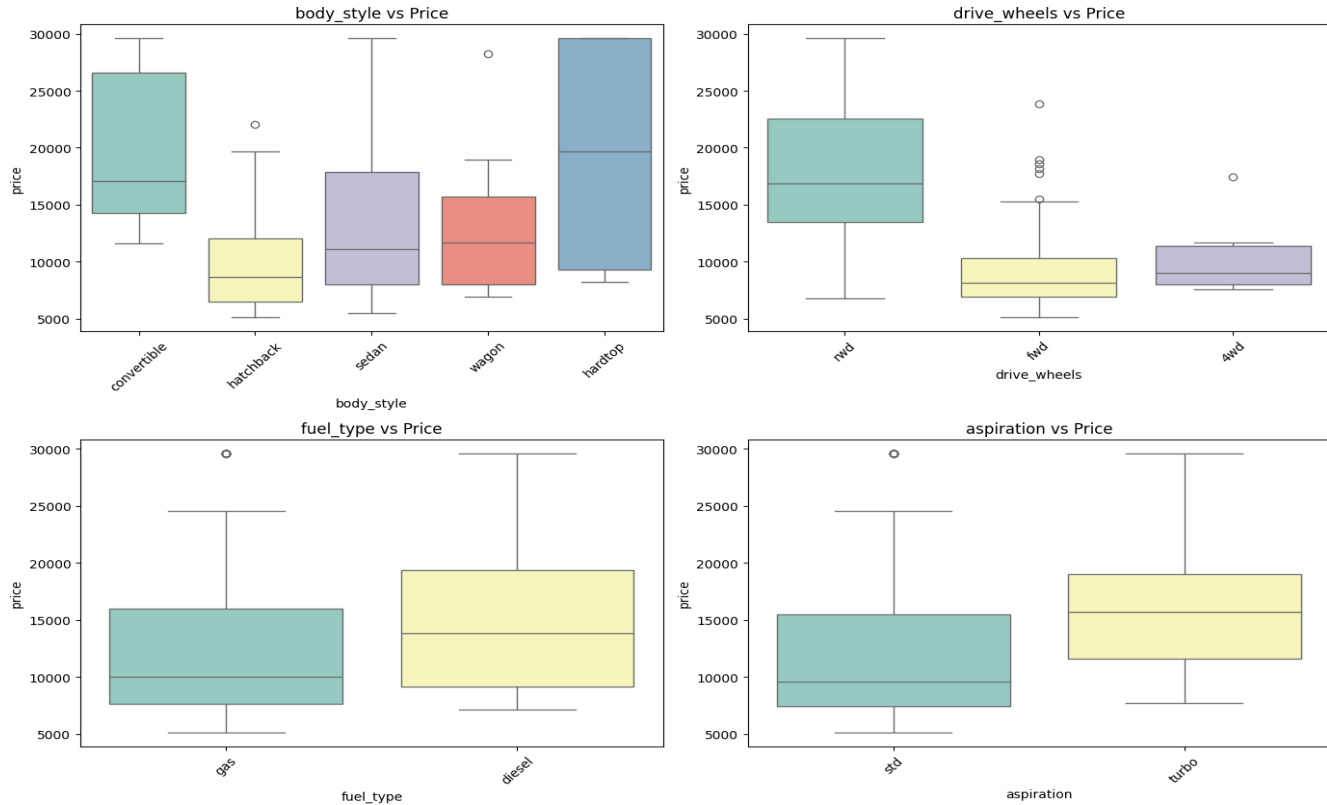


Figure 3. Price distribution by body style, drive wheels, fuel type, and aspiration.

4. Data Cleaning & Feature Engineering

- Column names supplied manually (raw file ships with no header row).
- "?" placeholder values converted to NaN via `na_values='?'` at load time.
- `num_of_doors` / `num_of_cylinders`: word-to-integer mapping (e.g. "four" → 4) to preserve ordinality instead of one-hot encoding, applied before splitting into X and y.
- Outlier capping via $1.5 \times \text{IQR}$ clipping across numeric columns, to reduce the influence of extreme values without discarding rows.
- Engineered features: power-to-weight ratio, engine size per cylinder, average MPG, body volume, length-to-wheelbase ratio, and horsepower per unit of engine size.
- ColumnTransformer pipeline: median imputation + standard scaling for numeric features; most-frequent imputation + one-hot encoding for nominal categorical features.
- 80/20 train/test split, identical pipeline applied to every candidate model for a fair comparison.

5. Modeling & Results

Ten algorithms were evaluated: Linear Regression, Ridge, Lasso, ElasticNet, KNN, SVR, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Each was scored with 5-fold cross-validated R^2 (primary metric) plus test R^2 , MAE, and RMSE as a sanity check.

Model	Train R^2	Test R^2	CV R^2 (mean)	Test MAE (\$)	Test RMSE (\$)
Gradient Boosting	0.998	0.974	0.874	991	1,434
Ridge	0.954	0.963	0.855	1,390	1,706
Linear Regression	0.965	0.969	0.853	1,163	1,570
Random Forest	0.982	0.957	0.853	1,271	1,844
Lasso	0.965	0.970	0.849	1,143	1,545
XGBoost	0.999	0.976	0.840	892	1,378
ElasticNet	0.846	0.793	0.806	2,960	4,045
SVR	0.970	0.930	0.804	1,709	2,350
Decision Tree	0.960	0.958	0.757	1,204	1,814
KNN	0.850	0.749	0.730	2,848	4,458

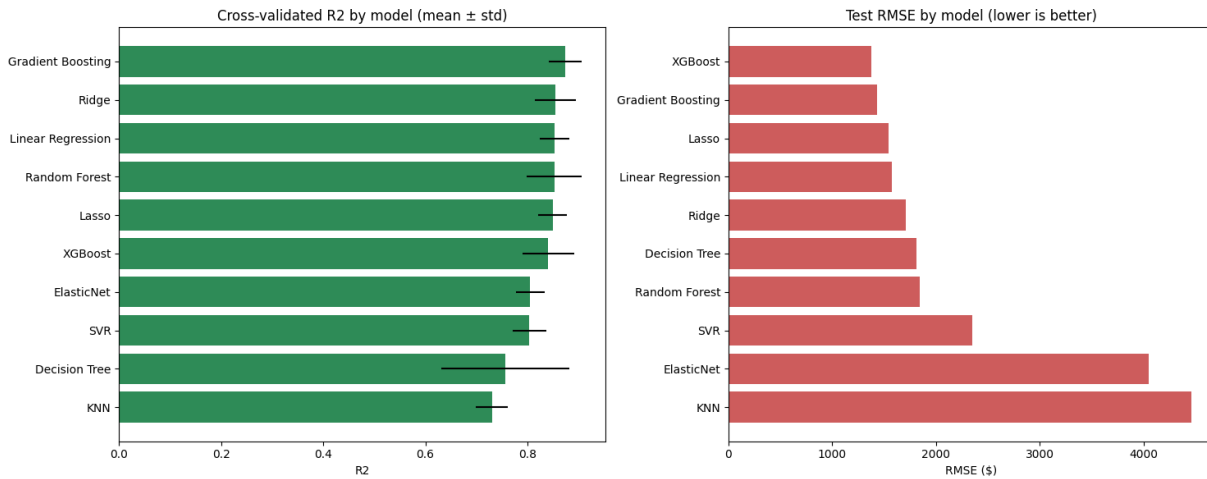


Figure 4. Cross-validated R^2 (left) and test RMSE (right) by model.

5.1 Best model: Gradient Boosting

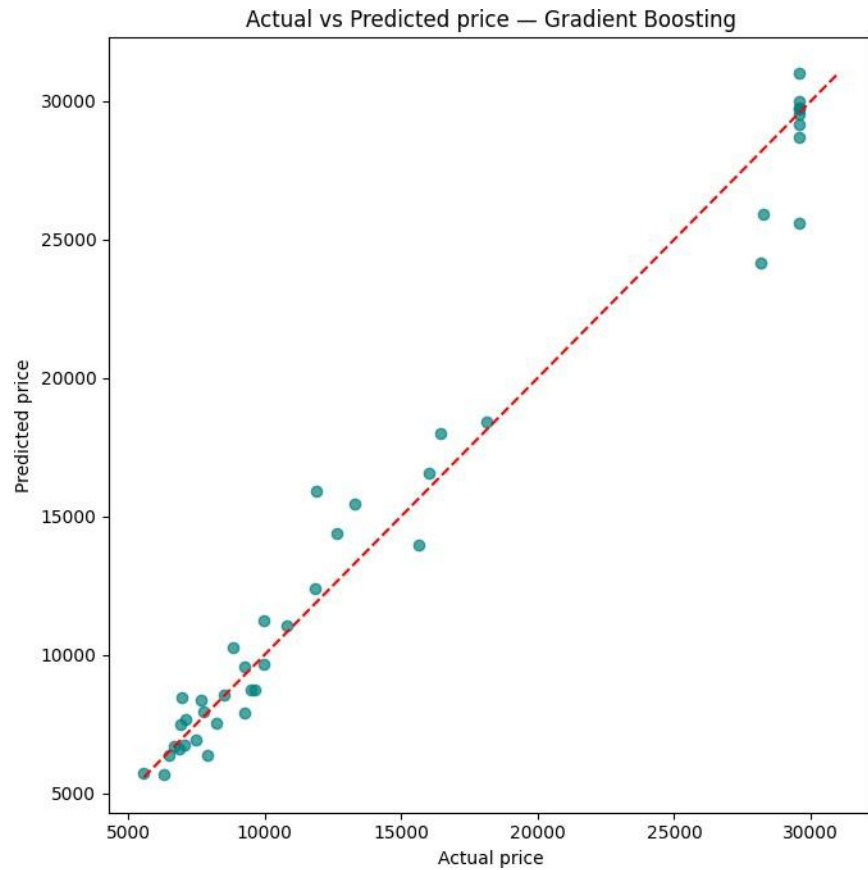


Figure 5. Actual vs. predicted price on the held-out test set.

5.2 Hyperparameter tuning

GridSearchCV over `n_estimators`, `learning_rate`, and `max_depth` selected `{n_estimators: 200, learning_rate: 0.05, max_depth: 3}`:

- Tuned CV R^2 : 0.879 (up from 0.874 untuned)
- Tuned test R^2 : 0.976 | Test MAE: \$988.70 | Test RMSE: \$1,383.02

The tuned model uses shallower trees than a naively deep default, trading a small amount of training-set fit for better-regularized, more generalizable predictions.

6. Challenges Faced

- No header row in the raw CSV — resolved by supplying column names explicitly.
- Missing values encoded as "?" — resolved with `na_values="?"` at load time.
- Missing predictor values — handled via pipeline imputation (median / most-frequent).
- Ordinal text fields (`num_of_doors`, `num_of_cylinders`) — mapped to integers to preserve order instead of one-hot encoding.
- Stale column-list variables computed during EDA (on the full dataframe, including price) were mistakenly reused when building the `ColumnTransformer`, raising `ValueError: A given column is not a column of the dataframe` — resolved by recomputing feature lists from `X` after the target was dropped.
- Skewed target and several skewed numeric features — addressed with $1.5 \times \text{IQR}$ clipping rather than discarding rows.
- High-cardinality, imbalanced make column (22 categories) — one-hot encoded, monitored via feature importance rather than dropped.
- Multicollinearity among size/engine features — addressed via regularization (Ridge/Lasso/ElasticNet) and by favoring ensembles that are robust to correlated inputs.
- Small test split (41 rows) is noisy — 5-fold CV R^2 used as the primary, more stable ranking metric.
- Model persistence risk — a pickled scikit-learn/XGBoost pipeline can fail to load under a different library version; mitigated by saving the full pipeline as one artifact and pinning versions between save and load environments.

7. Conclusion & Recommendation

Engine size, curb weight, horsepower, and body dimensions are confirmed as the primary positive drivers of automobile price, while fuel-economy metrics are the primary negative drivers; make and body style contribute additional brand/segment pricing power.

The tuned Gradient Boosting model is recommended for production use ($\text{CV } R^2 = 0.879$, $\text{test } R^2 = 0.976$). Ridge regression is retained as a secondary, fully interpretable model: its coefficients give stakeholders a transparent view of how each specification change is expected to move price, complementing the higher-accuracy but less transparent ensemble model.